

Does Cognitive Workload Impact the Effect on Pain Distraction in Virtual Reality? A Study Design

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ABSTRACT

Pain is an important issue for post-surgery patients. Virtual reality treatment (VRT) has been used to reduce pain perception, while there are few studies have investigated the effect of cognitive workload (CWL). The present study design hypothesizes that higher CWL will lead to stronger distraction in a virtual environment. To this end, a soothing psychological relief environment with a cognitive task, in which participants' CWL can be controlled, is constructed. In the future, this study will be applied to patients with pain to explore whether CWL is one of the key factors in reducing pain in VRT.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Pain is an important problem for patients after surgery. Conventional pharmacological treatment to deal with postoperative pain is using opioids. However, the use of opioids may lead to a variety of negative effects such as gastrointestinal and endocrine dysfunction, and even cause delirium in patients. Non-pharmacological interventions (NPI) do not produce side effects associated with pharmacological interventions and seem to be potentially safer methods. The effectiveness of NPI has been explored and studied in various medical subdivisions.

According to the gate-control theory (GCT) of pain [4], pain perception is influenced by a combination of sensory, behavioral, and psychological factors. Therefore, by increasing the amount of information that people have to process allows them to engage more in higher-order thinking [7]. Based on the multiple resource theory (MRT) [6], the cognitive resources of humans are limited. When additional CWL is provided to the user through tasks and environment, the competition for resources makes people's attention distracted so that pain perception is reduced. [1].

VRT has been increasingly used as an NPI for pain reduction. However, the exact mechanisms are unknown [7]. The present study conjectures that the distraction effect of users in virtual environments (VEs) is positively correlated with CWL, i.e., higher CWL lead to stronger distraction. Based on the above assumptions, a natural virtual environment incorporating a task to control CWL is designed and developed. We will apply our system to postoperative pain patients to investigate the relationship between CWL and pain distraction effects in this VE.

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2 STUDY DESIGN

2.1 participants

Target participants of this study are postoperative pain patients in hospitals. Sample size is calculated by Gpower 3.1. At a moderate effect level ($\alpha=0.05$, power=0.8), 168 participants need to be recruited. The actual number of recruits will be 210 based on an 80% completion rate prediction. Participants are over 18 years old, able to read Chinese, and able to move their head and hands freely will be included in the study. Participants with visual or auditory impairment, with cognitive or psychological impairment, epilepsy, and without persistent pain will be excluded from the experiment. Participants will be asked to drop out if they refuse to continue or the participant's health condition does not allow to continue participating.

2.2 Interventions

In this study, participants will be randomly assigned to three groups: a control group in which participants will receive regular medical care, an immersion group in which participants will experience VE without cognitive task, and an intervention group in which participants will experience VE and require to perform cognitive task that increase their CWL. This study implements a single-blind experiment because of the differences between interventions and hardware in each subgroup. The randomization process will be performed by healthcare professionals who are not related to the study.

2.3 Happy Valley Virtual Environment

The VE of Happy Valley designed in this paper is shown in Fig. 1. Happy Valley is developed based on the Unreal Engine. It allows users to move within the environment by walking and teleporting. A cognitive task is included in the VE to control participants' CWL.

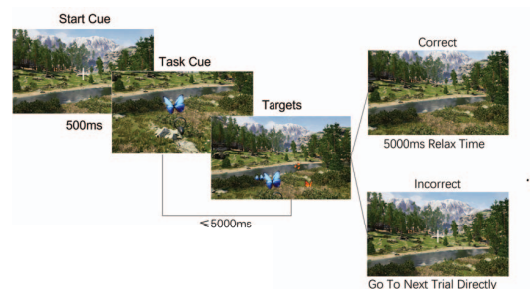


Figure 1: Screenshots of Happy Valley and the workflow of a single trial in the cognitive task.

2.4 Cognitive Task Design

The cognitive task is shown in Fig. 1. At the beginning of each trial, participants will see a crosshair in the center of their visual field that lasts 500ms, indicating that their task is about to start.

The task cue will then appear above the participants' controller. At the same time, a manageable number of task targets appear at random locations within a 120° horizontal field of view directly in front of the participants and within a distance of 3m to 15m. The participant needs to determine whether the task targets contain the same number of butterflies but different color with the task cue. This process needs to be completed within 5000ms. The participants can make selection by using the rocker of any controller, push up means participants answer "yes", while pull down means they answer "no" to the task. After the participants finish the selection, the system will judge whether the current selection is correct. If it is correct, the participants can get 5000ms of task-free relaxation time. Otherwise, the participants will start the next trial directly.

2.5 Measurements

2.5.1 Demographic survey

During the recruitment process, demographic surveys are conducted on the participants' gender, age, type of surgery, medication use, VR use experience, and video game experience.

2.5.2 Task performance

The number of trials performed by the participants in the whole experimental process and the correct rate will be recorded and calculated automatically by the system.

2.5.3 Objective indicators

This includes participants' (1) physiological recording (heart rate, respiratory rate, and oxygen saturation) via monitoring instruments, and (2) the amount of pain medication used by nurses.

2.5.4 Subjective indicators

This includes participants' (1) subjective workload measured by the subjective workload assessment technique (SWAT) [5], (2) presence in virtual reality by immersive tendencies questionnaire (ITQ) [9], (3) pain intensity before and after intervention by the Visual Analogue Scale (VAS) [2], (4) motion sickness symptoms after using the system by the simulator sickness questionnaire (SSQ) [3], (5) emotion states by the positive and negative affect survey (PANAS) [8], (6) investigating other aspects through self-administered scales (for example they encountered difficulties during use, whether they felt immersed in the environment, and whether they felt more physically comfortable during use compared to usual).

2.6 Study Procedure

This study consist of a test that will last three consecutive days. Participants will be surveyed demographically during the recruitment process. After confirming that participants meet the participation criteria, pain status, physiological indicators will be measured as baseline status. The intervention will be administered at a fixed time each day. Before conducting the intervention each day, the participants' SSQ, PANAS and pain status (VAS) will be measured. During the intervention, the physiological indicators will be continuously recorded by monitoring instrument, while the task performance will be continuously recorded by the VR system. After completing the intervention, all subjective indicators of the participants will be measured except for the self-made scale. After the 3-days intervention completed, the self-made scale will be filled in by the participants.

3 DISCUSSION

The study design used a randomized controlled experiment to compare three modalities in a single-blind situation. In the no-intervention case, participants only undergo the conventional treatment intervention without additional interventions, which is the same as the treatment process that everyday patients receive in the hospital. The VR immersion situation is similar to the traditional VR distraction process, where the user is immersed in a constructed

3D relaxing VE. By comparing the above two interventions, it is possible to study the distraction provided by the immersion environment. The VR intervention condition was added to the immersion condition with a cognitive task, causing the users in the intervention group to have an additional CWL from the task. During the cognitive task, participants need to complete the visual search, matching, and memory processing. Therefore, they need to engage more attention. The effects of the increased CWL will be investigated by comparing the intervention group with the immersion group.

The application of VRT to relief various types of pain has been increasingly studied. But the specific mechanisms of VRT are unclear. This study investigates the role of CWL on pain distraction in a customized VE called Happy Valley. A cognitive task in VR is designed and implemented within the virtual environment. The user study will be conducted in medical facilities under the guidance of medical staffs. This study are expected to provide new ideas for the design of future VR pain distraction systems.

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